DecoR: deconfounding time series with robust regression [4]

Joon Kim

Seminar "Advances in Statistical Inferences"

May 21, 2026

Keywords: Causal Inference · Robust Regression · Time Series Analysis

Outline

Outline

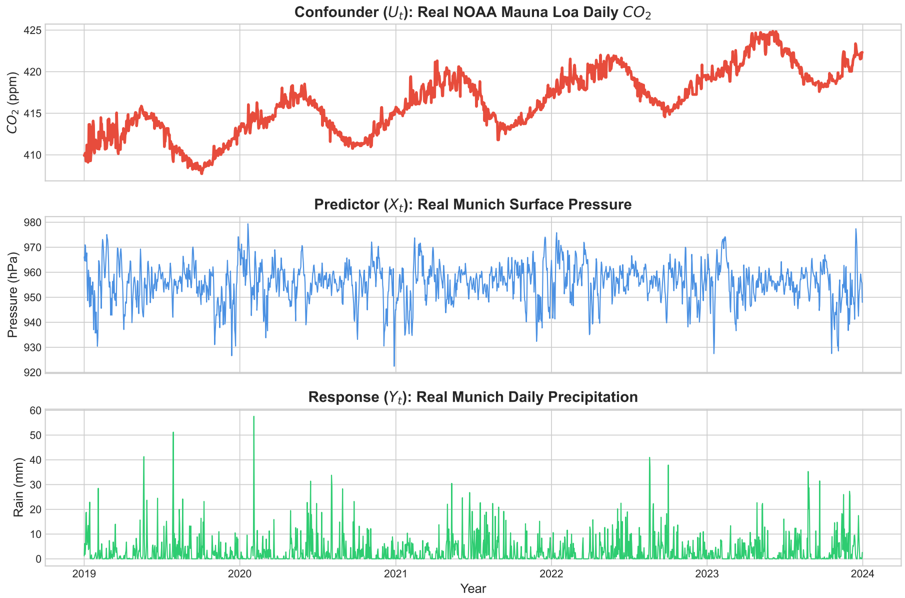


Figure: Real climate data fetched from Open Meteo [6] and Mauna Loa Observatory [2]

Importance of Decounfounding - Earth Science

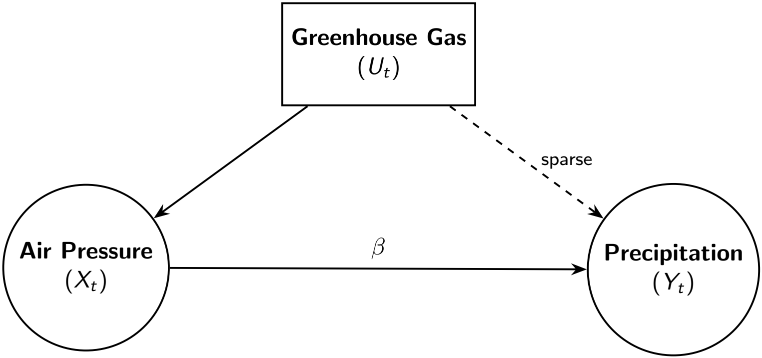


Figure: Causal Graph of Greenhouse Gas Confounding [5]

- **Goal:** Estimate β
- **Problem:** Biased estimation of β via OLS due to **hidden** confounder U
- **Idea:** Remove U, then estimate β

Outline

Data Generating Process (Continuous)

Let $T \in \mathbb{R}_{>0}$ be a fixed time horizon and $d \in \mathbb{N}$. We consider the relationship:

$$Y_t^{cont} = X_t^{cont}\beta + U_t^{cont} + \epsilon_t, \quad t \in [0, T]$$

Components:

- $Y_t^{cont} \in \mathbb{R}$, $X_t^{cont} \in \mathbb{R}^d$, $U_t^{cont} \in \mathbb{R}$: (continuous) stochastic processes.
- ϵ_t: i.i.d. Gaussian noise with variance σ.

In pratice: We can only observe discrete points.

Data Generating Process (Discrete)

Discretization:

Given a fixed $n \in \mathbb{N}$, we discretize with $\Delta t = T/n$:

- $X := (X^{cont}_{\Delta t}, X^{cont}_{2\Delta t}, ..., X^{cont}_{n\Delta t})^\top \in \mathbb{R}^{n \times d}$
- $Y := (Y^{cont}_{\Delta t}, Y^{cont}_{2\Delta t}, ..., Y^{cont}_{n\Delta t})^\top \in \mathbb{R}^{n}$
- $U := (U^{cont}_{\Delta t}, U^{cont}_{2\Delta t}, ..., U^{cont}_{n\Delta t})^\top \in \mathbb{R}^{n}$
- $\epsilon := (\epsilon^{cont}_{\Delta t}, \epsilon^{cont}_{2\Delta t}, ..., \epsilon^{cont}_{n\Delta t}) \in \mathbb{R}^{n}$

Our relationship still holds in the discrete world:

$$Y = X\beta + U + \epsilon$$

Observations:
We can only observe X and Y (and not hidden U and noise ϵ).

Frequency Domain Transformation

Define FDT as a function T, linearly mapping $\mathbb{R}^{n\times d}$ to $\mathbb{R}^{n\times d}$:

$$T(X) = \mathbf{T}X \quad , \text{with } \mathbf{T} \in \mathbb{R}^{n\times n}$$

Filling $\mathbf{T}$ with (orthonormal) basis functions $\phi_k : [0, T] \rightarrow \mathbb{R}$,
$k \in \{0, 1, ..., n-1\} :=$ frequency index of a basis function:

$$\mathbf{T} = \frac{1}{n} \underbrace{\begin{bmatrix} \phi_0(\Delta t) & \phi_1(\Delta t) & \dots & \phi_{n-1}(\Delta t) \\ \phi_0(2\Delta t) & \phi_1(2\Delta t) & \dots & \phi_{n-1}(2\Delta t) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_0(n\Delta t) & \phi_1(n\Delta t) & \dots & \phi_{n-1}(n\Delta t) \end{bmatrix}^\top}_{\mathbf{\Phi}}$$

$\mathbf{\Phi}$ is an orthogonal matrix. We will use an (orthonormal) cosine basis.

Frequency Domain Transformation

Define FDT as a function T, linearly mapping $\mathbb{R}^{n\times d}$ to $\mathbb{R}^{n\times d}$:

$$T(X) = \mathbf{T}X \quad \text{, with } \mathbf{T} \in \mathbb{R}^{n\times n}$$

Filling $\mathbf{T}$ with (orthonormal) basis functions $\phi_k : [0, T] \to \mathbb{R}$,
$k \in \{0, 1, ..., n-1\} :=$ frequency index of a basis function:

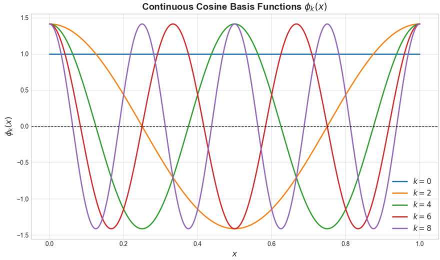
$$\mathbf{T} = \frac{1}{n}\underbrace{\begin{bmatrix} \phi_0(\Delta t) & \phi_1(\Delta t) & \dots & \phi_{n-1}(\Delta t) \\ \phi_0(2\Delta t) & \phi_1(2\Delta t) & \dots & \phi_{n-1}(2\Delta t) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_0(n\Delta t) & \phi_1(n\Delta t) & \dots & \phi_{n-1}(n\Delta t) \end{bmatrix}^{\top}}_{\mathbf{\Phi}}$$

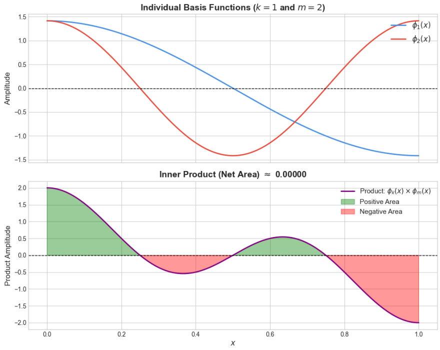
$\mathbf{\Phi}$ is an orthogonal matrix. We will use an (orthonormal) cosine basis.

Frequency Domain Transformation (Cosine Basis)

Define (orthonormal) cosine basis function $\phi_k : [0, T] \to \mathbb{R}$ by:

$$t \mapsto \begin{cases} 1 & \text{if } k = 0 \text{ and} \\ \sqrt{2}\cos(t\pi(k+1/2)) & \text{otherwise.} \end{cases}$$





Assumptions and Sparsity

Define:

- $G \subseteq \{0, ..., n-1\}$:= **small** set of frequency indices.
- E.g. $|G| \leq c = \frac{1}{10}n$.

- $U := (U^{cont}_{\Delta t}, U^{cont}_{2\Delta t}, ..., U^{cont}_{n\Delta t})^\top \in \mathbb{R}^n$.

Then, $T(U) \in \mathbb{R}^n$ is assumed to be *sparse*:

$$T(U)_k = 0 \quad \forall k \notin G$$

U is acting only on a few cosine basis frequencies.

Example: Sparse U

```python
import numpy as np
import math
from scipy.fft import dct

n = 50
T = 1.0  # Time horizon
t = np.linspace(0, T, n, endpoint=False)

# some small frequency set G
k1, k2 = 2, 5

# construct U only out of two frequencies
U_time = 2.0 * math.sqrt(2) * np.cos(2 * np.pi * t * (k1 + 1/2)) + 1.0 * math.sqrt(2) * np.cos(2 * np.pi * t * (k2 + 1/2))

# T(U)
U_freq = dct(U_time, type=3)
magnitude = np.abs(U_freq)

# 'U_freq' is zero everywhere except
# exactly at the integer indices k=2 and k=5
```

Example: Sparse U

```python
import numpy as np
import math
from scipy.fft import dct

n = 50
T = 1.0  # Time horizon
t = np.linspace(0, T, n, endpoint=False)

# some small frequency set G
k1, k2 = 2, 5

# construct U only out of two frequencies
U_time = 2.0 * math.sqrt(2) * np.cos(2 * np.pi * t * (k1 + 1/2)) +
    1.0 * math.sqrt(2) * np.cos(2 * np.pi * t * (k2 + 1/2))

# T(U)
U_freq = dct(U_time, type=3)
magnitude = np.abs(U_freq)

# 'U_freq' is zero everywhere except
# exactly at the integer indices k=2 and k=5

```

Example: Sparse U

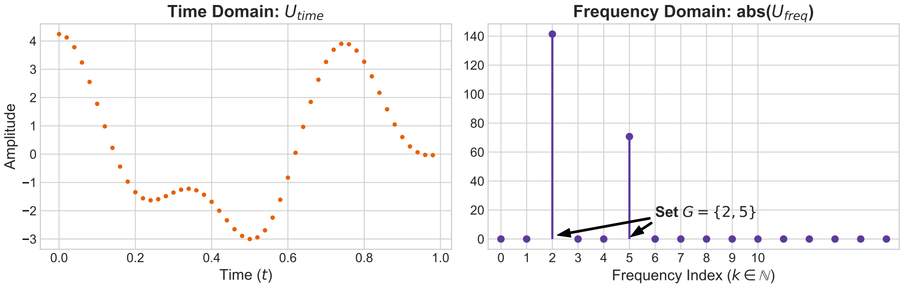


Figure: U in time domain and $T(U)$ in frequency domain.

The non-zero spikes represent the "small" confounded set G.

Frequency Domain Transformation - Outlier Problem

from before:

$$Y = X\beta + U + \epsilon$$

Using definition of $T(X) = \mathbf{T}X$:

$$T(Y) = T(X\beta + U + \epsilon) = T(X)\beta + T(U) + T(\epsilon)$$

The Deconfounded View

For k-th entry in $T(Y)$:

$$T(Y)_k \overset{\text{a.s.}}{=} \begin{cases} (T(X)\beta)_k + T(U)_k + T(\epsilon)_k & \text{if } k \in G, \\ (T(X)\beta)_k \qquad\qquad\quad + T(\epsilon)_k & \text{if } k \notin G. \end{cases}$$

Idea: The Outlier Problem

Consider transformed pairs $\{(T(X)_k, T(Y)_k\}_{k \le n}$:

If $k \in G \implies (T(X)_k, T(Y)_k)$ act as **outliers**.

If $k \notin G \implies$ estimate β via OLS.

Frequency Domain Transformation - Outlier Problem

from before:

$$Y = X\beta + U + \epsilon$$

Using definition of $T(X) = \mathbf{T}X$:

$$T(Y) = T(X\beta + U + \epsilon) = T(X)\beta + T(U) + T(\epsilon)$$

The Deconfounded View

For k-th entry in $T(Y)$:

$$T(Y)_k \overset{\text{a.s.}}{=} \begin{cases} (T(X)\beta)_k + T(U)_k + T(\epsilon)_k & \text{if } k \in G, \\ (T(X)\beta)_k \qquad\qquad\quad + T(\epsilon)_k & \text{if } k \notin G. \end{cases}$$

Idea: The Outlier Problem

Consider transformed pairs $\{(T(X)_k, T(Y)_k\}_{k \leq n}$:

If $k \in G \quad \Longrightarrow \quad (T(X)_k, T(Y)_k)$ act as **outliers**.

If $k \notin G \quad \Longrightarrow \quad$ estimate β via OLS.

Frequency Domain Transformation - Outlier Problem

from before:

$$Y = X\beta + U + \epsilon$$

Using definition of $T(X) = \mathbf{T}X$:

$$T(Y) = T(X\beta + U + \epsilon) = T(X)\beta + T(U) + T(\epsilon)$$

The Deconfounded View

For k-th entry in $T(Y)$:

$$T(Y)_k \overset{\text{a.s.}}{=} \begin{cases} (T(X)\beta)_k + T(U)_k + T(\epsilon)_k & \text{if } k \in G, \\ (T(X)\beta)_k \qquad\qquad\quad + T(\epsilon)_k & \text{if } k \notin G. \end{cases}$$

Idea: The Outlier Problem

Consider transformed pairs $\{(T(X)_k, T(Y)_k\}_{k \leq n}$:

If $k \in G \quad \Longrightarrow \quad (T(X)_k, T(Y)_k)$ act as **outliers**.

If $k \notin G \quad \Longrightarrow \quad$ estimate β via OLS.

Outlier Problem

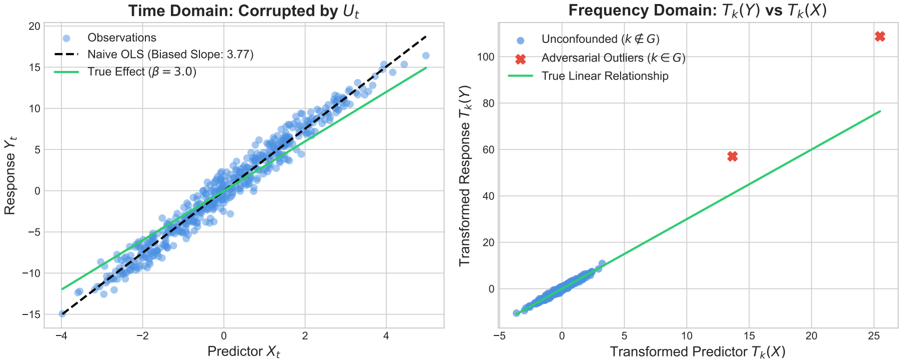


Figure: In time domain, U corrupts X (and the estimation of β via OLS). Applying $T(\cdot)$ isolates the confounder as outliers. Here, $X \sim N(0,1)$

Goal: identifying these outliers and estimate the true β (via Robust Regression).

Outline

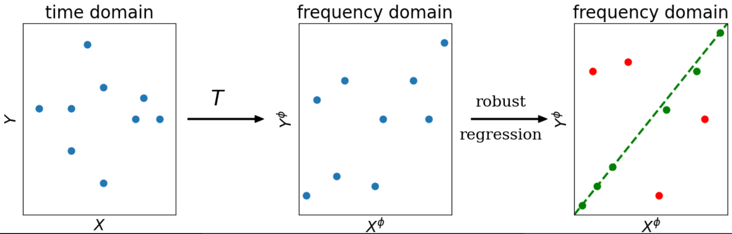
The DeCoR Framework

Require: $X \in \mathbb{R}^{n \times d}$, $Y \in \mathbb{R}^n$,
Transformation function T, robust regression algorithm $\mathcal{A}$

$X^\phi \leftarrow T(X)$ ▷ Translate to frequency domain
$Y^\phi \leftarrow T(Y)$

$\hat{\beta}_{\text{DeCoR}} \leftarrow \mathcal{A}(X^\phi, Y^\phi)$ ▷ robust regression

return $\hat{\beta}_{\text{DeCoR}}$



Torrent Algorithm [1]

Require $X^{\phi} \in \mathbb{R}^{n \times d}$, $Y^{\phi} \in \mathbb{R}^{n}$, $a := (n - c) \leq |G^{c}|$

Initialize: $S_0 \leftarrow \{1, \ldots, n\}, \text{currErr} \leftarrow \infty, t \leftarrow 0$

do

$\hat{\beta} \leftarrow \hat{\beta}_{\text{OLS}}(X^{\phi}_{S_t}, Y^{\phi}_{S_t})$ ▷ Fit OLS on current S_t

$v \leftarrow |Y^{\phi} - X^{\phi}\hat{\beta}|$ ▷ Calc residuals for ALL data points

$t \leftarrow t + 1$

$S_t \leftarrow \text{HT}(v, a)$ ▷ Keep a points with smallest residuals

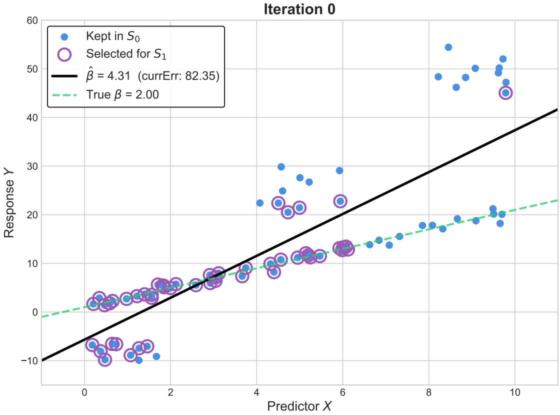
$\text{lastErr} \leftarrow \text{currErr}$

$\text{currErr} \leftarrow \|Y^{\phi}_{S_t} - X^{\phi}_{S_t}\hat{\beta}\|_2$ ▷ Check whether we improved or not

while $\text{currErr} < \text{lastErr}$

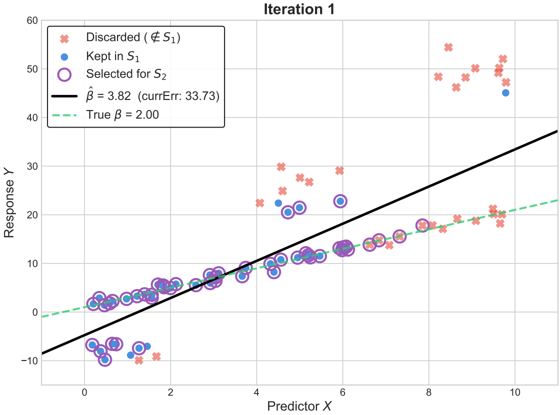
return $\hat{\beta}^{t}_{\text{Tor}}$

Torrent Algorithm [1] - Iteration-wise Visualization



Using $n = 80, a = 50, c = 30$

Start with:
$t = 0, \text{currErr} = \infty$

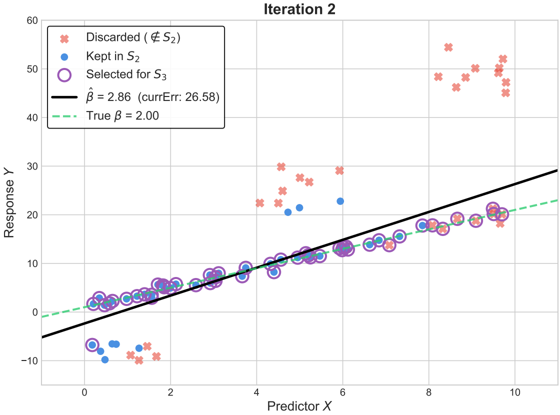


Start with:

$t = 1, \text{currErr} = 82.35$

Using $n = 80, a = 50, c = 30$

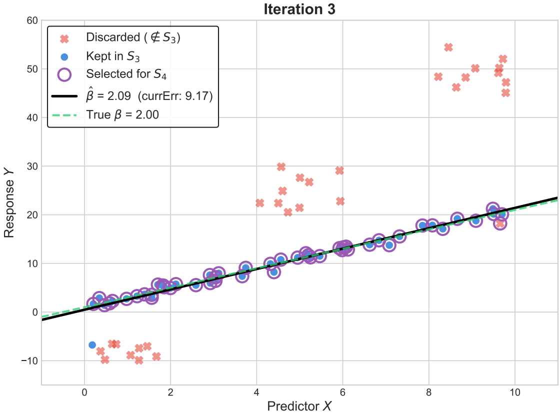
Torrent Algorithm [1] - Iteration-wise Visualization



Using $n = 80, a = 50, c = 30$

Start with:
$t = 2, \text{currErr} = 33.73$

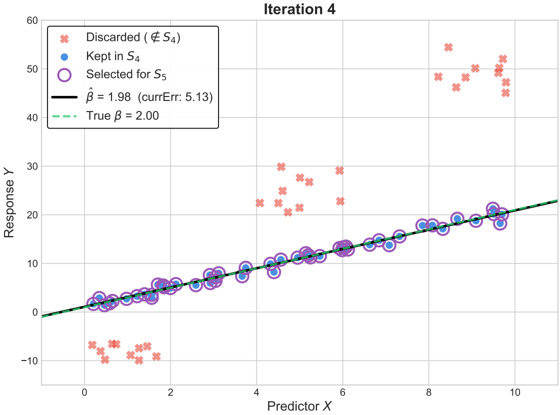
Torrent Algorithm [1] - Iteration-wise Visualization



Using $n = 80, a = 50, c = 30$

Start with:
$t = 3, \mathrm{currErr} = 26.58$

Torrent Algorithm [1] - Iteration-wise Visualization



Using $n = 80, a = 50, c = 30$

Start with:

$t = 4, \mathrm{currErr} = 9.17$

Outline

Convergence properties of DecoR-Tor (Theorem 6)

$$\|\hat{\beta}_{\text{DecoR}} - \beta\|_2 \in \mathcal{O}_{\mathbb{P}}\left(\sigma\sqrt{\frac{c_n \log(n/c_n)}{n}}\right).$$

Insights:

- $\hat{\beta}_{\text{DecoR}}$ is a consistent estimator of β, if $\sigma = 0$ or $c_n \in o(n)$
- if $\sigma \neq 0$ and c_n asymptotically proportional to n, then $\|\hat{\beta}_{\text{DecoR}} - \beta\|_2$ is bounded by σ, up to constant factors.

Guarantee of Decor-Tor - Proof Structure

(Deterministic) Guarantee of Torrent (Theorem 3)

Let S_t be the final subset of Torrent algorithm on data X^ϕ, Y^ϕ. Then:

$$\|\hat{\beta}_{\text{Tor}} - \beta\|_2 \leq \text{Err}_{\text{OLS}} + \frac{\sqrt{2}\|X^\phi_{(S_t \triangle G^c)}\|_2 \cdot \text{Err}_{\text{OLS}} + \sqrt{2}\|\epsilon^\phi_{(S_t \triangle G^c)}\|_2}{\sqrt{\lambda_{\min}(X^{\phi\top}_{S_t} X^\phi_{S_t})}(1 - \sqrt{2}\eta)}.$$

with $\text{Err}_{\text{OLS}} = \|(X_{S_t}^\top X_{S_t})^{-1} X_{S_t}^\top \epsilon_{S_t}\|_2$ *being the baseline error of the clean subset.*

$\downarrow$ Frequency Transformation, Gaussian Concentration Bounds ...

(Probabilistic) Gaurantee of DecoR-Tor (Theorem 6)

(Deterministic) Guarantee of Torrent (Theorem 3)

Let S_t be the final subset of Torrent algorithm on data X^ϕ, Y^ϕ. Then:

$$\|\hat{\beta}_{\mathrm{Tor}} - \beta\|_2 \leq \mathrm{Err}_{\mathrm{OLS}} + \frac{\sqrt{2}\|X^\phi_{(S_t \triangle G^c)}\|_2 \cdot \mathrm{Err}_{\mathrm{OLS}} + \sqrt{2}\|\epsilon^\phi_{(S_t \triangle G^c)}\|_2}{\sqrt{\lambda_{\min}(X^{\phi\top}_{S_t} X^\phi_{S_t})}(1 - \sqrt{2}\eta)}.$$

with $\mathrm{Err}_{\mathrm{OLS}} = \|(X_{S_t}^\top X_{S_t})^{-1} X_{S_t}^\top \epsilon_{S_t}\|_2$ *being the baseline error of the clean subset.*

$\downarrow$ Frequency Transformation, Gaussian Concentration Bounds ...

(Probabilistic) Gaurantee of DecoR-Tor (Theorem 6)

Guarantee of Torrent (Theorem 3) - Proof Sketch

$$
\begin{aligned}
\|\hat{\beta}^t_{\text{Tor}} - \beta\|_2 &\leq \underbrace{\left\|({X^\phi}_{S_t}^\top {X^\phi}_{S_t})^{-1} {X^\phi}_{S_t}^\top {\epsilon^\phi}_{S_t}\right\|_2}_{\text{Err}_{\text{OLS}}} \\
&\quad + \left\|\left|({X^\phi}_{S_t}^\top {X^\phi}_{S_t})^{-1} {X^\phi}_{S_t}^\top\right|\right\|_2 \cdot \|{U^\phi}_{S_t}\|_2 \\
&= \text{Err}_{\text{OLS}} + \frac{1}{\sqrt{\lambda_{\min}({X^\phi}_{S_t}^\top {X^\phi}_{S_t})}} \cdot \|{U^\phi}_{S_t}\|_2 \\
&\leq \text{Err}_{\text{OLS}} + \frac{\sqrt{2}\left\|\left|{X^\phi}_{(S_t \triangle G^c)}\right|\right\|_2 \cdot \text{Err}_{\text{OLS}} + \sqrt{2}\|{\epsilon^\phi}_{(S_t \triangle G^c)}\|_2}{\sqrt{\lambda_{\min}({X^\phi}_{S_t}^\top {X^\phi}_{S_t})}(1-\sqrt{2}\eta)}
\end{aligned}
$$

We will prove: $\|U^\phi_{S_t}\|_2 \leq \dfrac{\sqrt{2}\left\|\left|{X^\phi}_{(S_t \triangle G^c)}\right|\right\|_2 \cdot \text{Error}_{\text{OLS}} + \sqrt{2}\|{\epsilon^\phi}_{(S_t \triangle G^c)}\|_2}{1-\sqrt{2}\eta}$

Guarantee of Torrent (Theorem 3) - (partial) Proof

Hard Thresholding Guarantee

Recall HT-step:

$$S_t \leftarrow \mathrm{HT}(v, a)$$

with $v :=$ vector of residuals for all data points, $a \leq |G^c|$

$$\begin{aligned}
\|X^{\phi}_{S_t}(\beta - \hat{\beta}^t_{\mathrm{Tor}}) + U^{\phi}_{S_t} + \epsilon^{\phi}_{S_t}\|_2^2 &= \|Y^{\phi}_{S_t} - X^{\phi}_{S_t}\hat{\beta}^t_{\mathrm{Tor}}\|_2^2 \\
&\leq \|Y^{\phi}_{G^c} - X^{\phi}_{G^c}\hat{\beta}^t_{\mathrm{Tor}}\|_2^2 \\
&= \|X^{\phi}_{G^c}(\beta - \hat{\beta}^t_{\mathrm{Tor}}) + \epsilon^{\phi}_{G^c}\|_2^2
\end{aligned}$$

Let $H_t := S_t \setminus G^c$ and $M_t := G^c \setminus S_t$.

We know from above:

$$\|X^{\phi}_{H_t}(\beta - \hat{\beta}^t_{\mathrm{Tor}}) + U^{\phi}_{H_t} + \epsilon^{\phi}_{H_t}\|_2 \leq \|X^{\phi}_{M_t}(\beta - \hat{\beta}^t_{\mathrm{Tor}}) + \epsilon^{\phi}_{M_t}\|_2$$

from before:

$$\|X^{\phi}_{H_t}(\beta - \hat{\beta}^t_{\text{Tor}}) + U^{\phi}_{H_t} + \epsilon^{\phi}_{H_t}\|_2 \leq \|X^{\phi}_{M_t}(\beta - \hat{\beta}^t_{\text{Tor}}) + \epsilon^{\phi}_{M_t}\|_2$$

Using $||z||_1 \leq \sqrt{2}||z||_2$ for a 2-dimensional vector z:

Outlier Expansion

$$\begin{aligned}
\|{U^{\phi}}_{S_t}\|_2 = \|{U^{\phi}}_{H_t}\|_2 & \\
&= \|{X^{\phi}}_{H_t}(\beta - \hat{\beta}^t_{\text{Tor}}) + U^{\phi}_{H_t} + \epsilon^{\phi}_{H_t} - ({X^{\phi}}_{H_t}(\beta - \hat{\beta}^t_{\text{Tor}}) + \epsilon^{\phi}_{H_t})\|_2 \\
&\leq \|{X^{\phi}}_{M_t}(\beta - \hat{\beta}^t_{\text{Tor}})\|_2 + \|{X^{\phi}}_{H_t}(\beta - \hat{\beta}^t_{\text{Tor}})\|_2 + \|\epsilon^{\phi}_{H_t}\|_2 + \|\epsilon^{\phi}_{M_t}\|_2 \\
&\leq \sqrt{2}\,\|{X^{\phi}}_{S_t \triangle G^c}\|_2\,\|\beta - \hat{\beta}^t_{\text{Tor}}\|_2 + \sqrt{2}\,\|\epsilon^{\phi}_{S_t \triangle G^c}\|_2 \\
&\leq \sqrt{2}\,\|{X^{\phi}}_{S_t \triangle G^c}\|_2 \left(\text{Err}_{\text{OLS}} + \left\|\left|({X^{\phi}}^{\top}_{S_t}{X^{\phi}}_{S_t})^{-1}{X^{\phi}}^{\top}_{S_t}\right|\right\|_2 \|{U^{\phi}}_{S_t}\|_2\right) \\
&\quad + \sqrt{2}\,\|\epsilon^{\phi}_{S_t \triangle G^c}\|_2 \\
&\leq \underbrace{\sqrt{2}\|{X^{\phi}}_{S_t \triangle G^c}\|_2 \left\|\left|\left({X^{\phi}}^{\top}_{S_t}{X^{\phi}}_{S_t}\right)^{-1}{X^{\phi}}^{\top}_{S_t}\right|\right\|_2}_{\leq \sqrt{2}\eta} \|{U^{\phi}}_{S_t}\|_2 \\
&\quad + \sqrt{2}\|{X^{\phi}}_{S_t \triangle G^c}\|_2 \cdot \text{Err}_{\text{OLS}} + \sqrt{2}\|\epsilon^{\phi}_{S_t \triangle G^c}\|_2
\end{aligned}$$

from before:

$$\|X^{\phi}_{H_t}(\beta - \hat{\beta}^t_{\text{Tor}}) + U^{\phi}_{H_t} + \epsilon^{\phi}_{H_t}\|_2 \le \|X^{\phi}_{M_t}(\beta - \hat{\beta}^t_{\text{Tor}}) + \epsilon^{\phi}_{M_t}\|_2$$

Using $||z||_1 \le \sqrt{2}||z||_2$ for a 2-dimensional vector z:

Outlier Expansion

$$\begin{aligned}
\|{U^{\phi}}_{S_t}\|_2 &= \|{U^{\phi}}_{H_t}\|_2 \\
&= \|{X^{\phi}}_{H_t}(\beta - \hat{\beta}^t_{\text{Tor}}) + U^{\phi}_{H_t} + \epsilon^{\phi}_{H_t} - ({X^{\phi}}_{H_t}(\beta - \hat{\beta}^t_{\text{Tor}}) + \epsilon^{\phi}_{H_t})\|_2 \\
&\le \|{X^{\phi}}_{M_t}(\beta - \hat{\beta}^t_{\text{Tor}})\|_2 + \|{X^{\phi}}_{H_t}(\beta - \hat{\beta}^t_{\text{Tor}})\|_2 + \|\epsilon^{\phi}_{H_t}\|_2 + \|\epsilon^{\phi}_{M_t}\|_2 \\
&\le \sqrt{2}\,\|{X^{\phi}}_{S_t \triangle G^c}\|_2\,\|\beta - \hat{\beta}^t_{\text{Tor}}\|_2 + \sqrt{2}\,\|\epsilon^{\phi}_{S_t \triangle G^c}\|_2 \\
&\le \sqrt{2}\,\|{X^{\phi}}_{S_t \triangle G^c}\|_2 \left(\text{Err}_{\text{OLS}} + \left\|\left|({X^{\phi}}^{\top}_{S_t}{X^{\phi}}_{S_t})^{-1}{X^{\phi}}^{\top}_{S_t}\right|\right\|_2 \|{U^{\phi}}_{S_t}\|_2\right) \\
&\quad + \sqrt{2}\,\|\epsilon^{\phi}_{S_t \triangle G^c}\|_2 \\
&\le \underbrace{\sqrt{2}\|{X^{\phi}}_{S_t \triangle G^c}\|_2 \left\|\left|\left({X^{\phi}}^{\top}_{S_t}{X^{\phi}}_{S_t}\right)^{-1}{X^{\phi}}^{\top}_{S_t}\right|\right\|_2}_{\le \sqrt{2}\eta} \|{U^{\phi}}_{S_t}\|_2 \\
&\quad + \sqrt{2}\|{X^{\phi}}_{S_t \triangle G^c}\|_2 \cdot \text{Err}_{\text{OLS}} + \sqrt{2}\|\epsilon^{\phi}_{S_t \triangle G^c}\|_2
\end{aligned}$$

from before:

$$\|X^{\phi}_{H_t}(\beta - \hat{\beta}^t_{\text{Tor}}) + U^{\phi}_{H_t} + \epsilon^{\phi}_{H_t}\|_2 \le \|X^{\phi}_{M_t}(\beta - \hat{\beta}^t_{\text{Tor}}) + \epsilon^{\phi}_{M_t}\|_2$$

Using $||z||_1 \le \sqrt{2}||z||_2$ for a 2-dimensional vector z:

Outlier Expansion

$$\begin{aligned}
\|{U^{\phi}}_{S_t}\|_2 &= \|{U^{\phi}}_{H_t}\|_2 \\
&= \|{X^{\phi}}_{H_t}(\beta - \hat{\beta}^t_{\text{Tor}}) + U^{\phi}_{H_t} + \epsilon^{\phi}_{H_t} - ({X^{\phi}}_{H_t}(\beta - \hat{\beta}^t_{\text{Tor}}) + \epsilon^{\phi}_{H_t})\|_2 \\
&\le \|{X^{\phi}}_{M_t}(\beta - \hat{\beta}^t_{\text{Tor}})\|_2 + \|{X^{\phi}}_{H_t}(\beta - \hat{\beta}^t_{\text{Tor}})\|_2 + \|\epsilon^{\phi}_{H_t}\|_2 + \|\epsilon^{\phi}_{M_t}\|_2 \\
&\le \sqrt{2}\,\|{X^{\phi}}_{S_t \triangle G^c}\|_2\,\|\beta - \hat{\beta}^t_{\text{Tor}}\|_2 + \sqrt{2}\,\|\epsilon^{\phi}_{S_t \triangle G^c}\|_2 \\
&\le \sqrt{2}\,\|{X^{\phi}}_{S_t \triangle G^c}\|_2 \left(\text{Err}_{\text{OLS}} + \left\|\left|({X^{\phi}}^{\top}_{S_t}{X^{\phi}}_{S_t})^{-1}{X^{\phi}}^{\top}_{S_t}\right\|\right|_2 \|{U^{\phi}}_{S_t}\|_2\right) \\
&\quad + \sqrt{2}\,\|\epsilon^{\phi}_{S_t \triangle G^c}\|_2 \\
&\le \underbrace{\sqrt{2}\|{X^{\phi}}_{S_t \triangle G^c}\|_2 \left\|\left|\left({X^{\phi}}^{\top}_{S_t}{X^{\phi}}_{S_t}\right)^{-1}{X^{\phi}}^{\top}_{S_t}\right\|\right|_2}_{\le \sqrt{2}\eta} \|{U^{\phi}}_{S_t}\|_2 \\
&\quad + \sqrt{2}\|{X^{\phi}}_{S_t \triangle G^c}\|_2 \cdot \text{Err}_{\text{OLS}} + \sqrt{2}\|\epsilon^{\phi}_{S_t \triangle G^c}\|_2
\end{aligned}$$

from before:

$$\|X^{\phi}_{H_t}(\beta - \hat{\beta}^t_{\text{Tor}}) + U^{\phi}_{H_t} + \epsilon^{\phi}_{H_t}\|_2 \leq \|X^{\phi}_{M_t}(\beta - \hat{\beta}^t_{\text{Tor}}) + \epsilon^{\phi}_{M_t}\|_2$$

Using $||z||_1 \leq \sqrt{2}||z||_2$ for a 2-dimensional vector z:

Outlier Expansion

$$\begin{aligned}
\|{U^{\phi}}_{S_t}\|_2 &= \|{U^{\phi}}_{H_t}\|_2 \\
&= \|{X^{\phi}}_{H_t}(\beta - \hat{\beta}^t_{\text{Tor}}) + U^{\phi}_{H_t} + \epsilon^{\phi}_{H_t} - ({X^{\phi}}_{H_t}(\beta - \hat{\beta}^t_{\text{Tor}}) + \epsilon^{\phi}_{H_t})\|_2 \\
&\leq \|{X^{\phi}}_{M_t}(\beta - \hat{\beta}^t_{\text{Tor}})\|_2 + \|{X^{\phi}}_{H_t}(\beta - \hat{\beta}^t_{\text{Tor}})\|_2 + \|\epsilon^{\phi}_{H_t}\|_2 + \|\epsilon^{\phi}_{M_t}\|_2 \\
&\leq \sqrt{2}\,\|{X^{\phi}}_{S_t \triangle G^c}\|_2\,\|\beta - \hat{\beta}^t_{\text{Tor}}\|_2 + \sqrt{2}\,\|\epsilon^{\phi}_{S_t \triangle G^c}\|_2 \\
&\leq \sqrt{2}\,\|{X^{\phi}}_{S_t \triangle G^c}\|_2 \left(\text{Err}_{\text{OLS}} + \left\|\left({{X^{\phi}}^{\top}_{S_t}}{X^{\phi}}_{S_t}\right)^{-1}{X^{\phi}}^{\top}_{S_t}\right\|_2 \|{U^{\phi}}_{S_t}\|_2\right) \\
&\quad + \sqrt{2}\,\|\epsilon^{\phi}_{S_t \triangle G^c}\|_2 \\
&\leq \underbrace{\sqrt{2}\|{X^{\phi}}_{S_t \triangle G^c}\|_2 \left\|\left({{X^{\phi}}^{\top}_{S_t}}{X^{\phi}}_{S_t}\right)^{-1}{X^{\phi}}^{\top}_{S_t}\right\|_2}_{\leq \sqrt{2}\eta} \|{U^{\phi}}_{S_t}\|_2 \\
&\quad + \sqrt{2}\|{X^{\phi}}_{S_t \triangle G^c}\|_2 \cdot \text{Err}_{\text{OLS}} + \sqrt{2}\|\epsilon^{\phi}_{S_t \triangle G^c}\|_2
\end{aligned}$$

from before:

$$\|X^{\phi}_{H_t}(\beta - \hat{\beta}^t_{\text{Tor}}) + U^{\phi}_{H_t} + \epsilon^{\phi}_{H_t}\|_2 \leq \|X^{\phi}_{M_t}(\beta - \hat{\beta}^t_{\text{Tor}}) + \epsilon^{\phi}_{M_t}\|_2$$

Using $||z||_1 \leq \sqrt{2}||z||_2$ for a 2-dimensional vector z:

Outlier Expansion

$$\begin{aligned}
\|{U^{\phi}}_{S_t}\|_2 &= \|{U^{\phi}}_{H_t}\|_2 \\
&= \|{X^{\phi}}_{H_t}(\beta - \hat{\beta}^t_{\text{Tor}}) + U^{\phi}_{H_t} + \epsilon^{\phi}_{H_t} - ({X^{\phi}}_{H_t}(\beta - \hat{\beta}^t_{\text{Tor}}) + \epsilon^{\phi}_{H_t})\|_2 \\
&\leq \|{X^{\phi}}_{M_t}(\beta - \hat{\beta}^t_{\text{Tor}})\|_2 + \|{X^{\phi}}_{H_t}(\beta - \hat{\beta}^t_{\text{Tor}})\|_2 + \|\epsilon^{\phi}_{H_t}\|_2 + \|\epsilon^{\phi}_{M_t}\|_2 \\
&\leq \sqrt{2}\,\|{X^{\phi}}_{S_t \triangle G^c}\|_2\,\|\beta - \hat{\beta}^t_{\text{Tor}}\|_2 + \sqrt{2}\,\|\epsilon^{\phi}_{S_t \triangle G^c}\|_2 \\
&\leq \sqrt{2}\,\|{X^{\phi}}_{S_t \triangle G^c}\|_2 \left(\text{Err}_{\text{OLS}} + \left\|\left|({X^{\phi}}^{\top}_{S_t}{X^{\phi}}_{S_t})^{-1}{X^{\phi}}^{\top}_{S_t}\right|\right\|_2 \|{U^{\phi}}_{S_t}\|_2\right) \\
&\quad + \sqrt{2}\,\|\epsilon^{\phi}_{S_t \triangle G^c}\|_2 \\
&\leq \underbrace{\sqrt{2}\|{X^{\phi}}_{S_t \triangle G^c}\|_2 \left\|\left|\left({X^{\phi}}^{\top}_{S_t}{X^{\phi}}_{S_t}\right)^{-1}{X^{\phi}}^{\top}_{S_t}\right|\right\|_2}_{\leq\sqrt{2}\eta} \|{U^{\phi}}_{S_t}\|_2 \\
&\quad + \sqrt{2}\|{X^{\phi}}_{S_t \triangle G^c}\|_2 \cdot \text{Err}_{\text{OLS}} + \sqrt{2}\|\epsilon^{\phi}_{S_t \triangle G^c}\|_2
\end{aligned}$$

from before:

$$\|X^{\phi}_{H_t}(\beta - \hat{\beta}^t_{\text{Tor}}) + U^{\phi}_{H_t} + \epsilon^{\phi}_{H_t}\|_2 \le \|X^{\phi}_{M_t}(\beta - \hat{\beta}^t_{\text{Tor}}) + \epsilon^{\phi}_{M_t}\|_2$$

Using $||z||_1 \le \sqrt{2}||z||_2$ for a 2-dimensional vector z:

Outlier Expansion

$$\begin{aligned}
\|{U^{\phi}}_{S_t}\|_2 &= \|{U^{\phi}}_{H_t}\|_2 \\
&= \|{X^{\phi}}_{H_t}(\beta - \hat{\beta}^t_{\text{Tor}}) + U^{\phi}_{H_t} + \epsilon^{\phi}_{H_t} - ({X^{\phi}}_{H_t}(\beta - \hat{\beta}^t_{\text{Tor}}) + \epsilon^{\phi}_{H_t})\|_2 \\
&\le \|{X^{\phi}}_{M_t}(\beta - \hat{\beta}^t_{\text{Tor}})\|_2 + \|{X^{\phi}}_{H_t}(\beta - \hat{\beta}^t_{\text{Tor}})\|_2 + \|\epsilon^{\phi}_{H_t}\|_2 + \|\epsilon^{\phi}_{M_t}\|_2 \\
&\le \sqrt{2}\,\|{X^{\phi}}_{S_t \triangle G^c}\|_2\,\|\beta - \hat{\beta}^t_{\text{Tor}}\|_2 + \sqrt{2}\,\|\epsilon^{\phi}_{S_t \triangle G^c}\|_2 \\
&\le \sqrt{2}\,\|{X^{\phi}}_{S_t \triangle G^c}\|_2 \left(\text{Err}_{\text{OLS}} + \left\|\left\|({X^{\phi}}^{\top}_{S_t}{X^{\phi}}_{S_t})^{-1}{X^{\phi}}^{\top}_{S_t}\right\|\right\|_2 \|{U^{\phi}}_{S_t}\|_2\right) \\
&\quad + \sqrt{2}\,\|\epsilon^{\phi}_{S_t \triangle G^c}\|_2 \\
&\le \underbrace{\sqrt{2}\|{X^{\phi}}_{S_t \triangle G^c}\|_2 \left\|\left\|\left({X^{\phi}}^{\top}_{S_t}{X^{\phi}}_{S_t}\right)^{-1}{X^{\phi}}^{\top}_{S_t}\right\|\right\|_2}_{\le\sqrt{2}\eta} \|{U^{\phi}}_{S_t}\|_2 \\
&\quad + \sqrt{2}\|{X^{\phi}}_{S_t \triangle G^c}\|_2 \cdot \text{Err}_{\text{OLS}} + \sqrt{2}\|\epsilon^{\phi}_{S_t \triangle G^c}\|_2
\end{aligned}$$

Continuing...

Outlier Expansion

$$\begin{aligned}\|{U^{\phi}}_{S_t}\|_2 &\leq \sqrt{2}\eta \cdot \|{U^{\phi}}_{S_t}\|_2 \\ &\quad + \sqrt{2}\|{X^{\phi}}_{S_t \triangle G^c}\|_2 \cdot \mathrm{Err}_{\mathrm{OLS}} + \sqrt{2}\|\epsilon^{\phi}_{S_t \triangle G^c}\|_2\end{aligned}$$

$$\implies \|{U^{\phi}}_{S_t}\|_2 \leq \frac{\sqrt{2}\|{X^{\phi}}_{S_t \triangle G^c}\|_2 \cdot \mathrm{Err}_{\mathrm{OLS}} + \sqrt{2}\|\epsilon^{\phi}_{S_t \triangle G^c}\|_2}{1 - \sqrt{2}\eta}$$

Continuing...

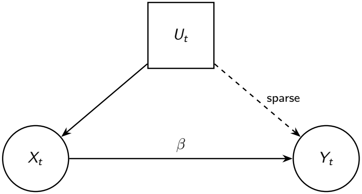
Outlier Expansion

$$\|U^{\phi}{}_{S_t}\|_2 \leq \sqrt{2}\eta \cdot \|U^{\phi}{}_{S_t}\|_2 + \sqrt{2}\|X^{\phi}{}_{S_t \triangle G^c}\|_2 \cdot \mathrm{Err}_{\mathrm{OLS}} + \sqrt{2}\|\epsilon^{\phi}_{S_t \triangle G^c}\|_2$$

$$\implies \|U^{\phi}{}_{S_t}\|_2 \leq \frac{\sqrt{2}\|X^{\phi}{}_{S_t \triangle G^c}\|_2 \cdot \mathrm{Err}_{\mathrm{OLS}} + \sqrt{2}\|\epsilon^{\phi}_{S_t \triangle G^c}\|_2}{1 - \sqrt{2}\eta}$$

Outline

Recall:



1-dimensional scenario:

- $U := \mathrm{OU}(1) + \epsilon_u$ with noise $\epsilon_u \sim N(0, 1)$
- U_{sparse} = Project U onto random G, $|G| = \frac{1}{4} \cdot n$
- $X := \mathrm{OU}(2) + U + \epsilon_x$ with $\epsilon_x \sim N(0, 1)$
- $Y := 3X + 10 \cdot U_{sparse} + \epsilon_y$ with $\epsilon_y \sim N(0, 1)$

Synthetic Data - Experimental Results

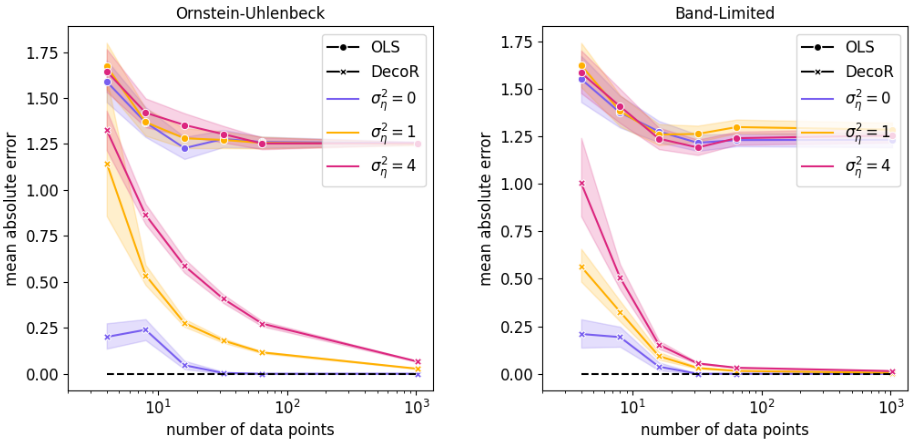


Figure: DecoR-Tor vs OLS with $|G_n| = \frac{1}{4}n$ and cosine-basis.
In the right, band B of size 50 is used.

Application: Earth System Science

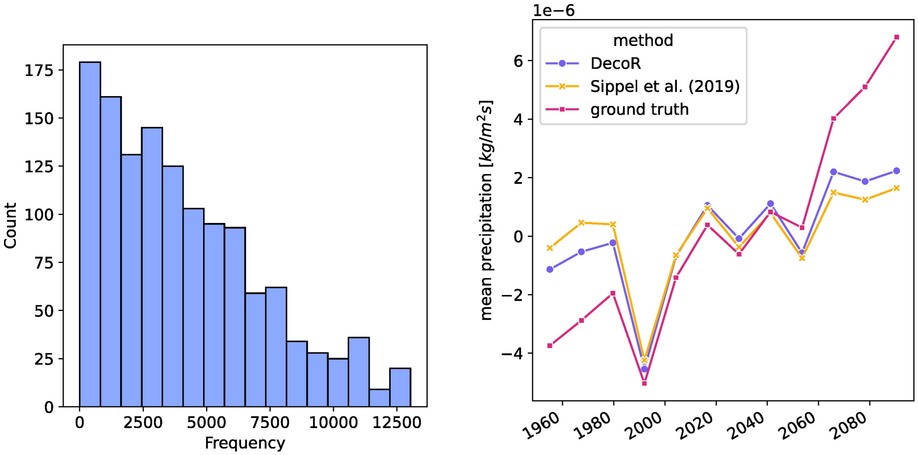


Figure: Frequencies filtered out by DecoR (left), 'ground truth' provided by ClimEx project[3]

Findings:

DeCoR-Tor filters out the low-frequency components: greenhouse gas.

Outline

Non-linear extension

Recall from before:

$$Y = X\beta + U + \epsilon$$

Consider non-linear relationship:

$$Y = f(X) + U + \epsilon$$

Under L^2 Assumption,
We approximate f as a linear combination of basis functions $\{\psi\}_{l\in\mathbb{N}}$:

$$\underbrace{\begin{bmatrix} f(X_{\Delta t}) \\ \vdots \\ f(X_{n\Delta t}) \end{bmatrix}}_{\mathbf{f}_{n\times 1}} \approx \underbrace{\begin{bmatrix} \psi_0(X_{\Delta t}) & \dots & \psi_L(X_{\Delta t}) \\ \vdots & \ddots & \vdots \\ \psi_0(X_{n\Delta t}) & \dots & \psi_L(X_{n\Delta t}) \end{bmatrix}}_{\mathbf{\Psi}_{n\times L}} \underbrace{\begin{bmatrix} \beta_0 \\ \vdots \\ \beta_{L-1} \end{bmatrix}}_{\beta_{L\times 1}}$$

Now, same setting as in linear case. Go ahead with DecoR-Tor.

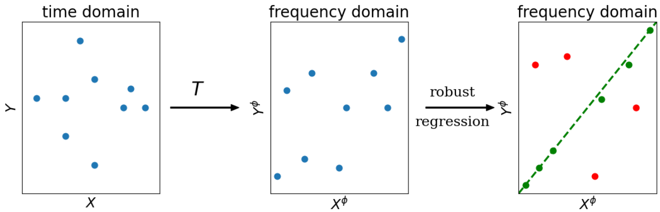
Outline

Conclusion

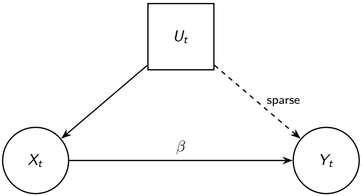
- **Problem:**
  Hidden confounders severely bias standard OLS.

- **Idea:**
  Transform into frequency domain, exploit sparsity of confounders.

- **Method:**
  - See confounded frequencies as adversarial outliers
  - Do robust regression (like Torrent)



Future Work



1. U **dense towards** Y **but sparse towards** X
2. **Irregular Intervals:** not sampled in equidistant timestamps
3. **Confidence Intervals:** Developing provably valid confidence intervals

Thank You!

Any questions?

Code used in this presentation:
github.com/rlawnsiiiii/robust_deconfounding

References I

Kush Bhatia, Prateek Jain, and Purushottam Kar.
Robust regression via hard thresholding, 2015.

Xin Lan and Ralph Keeling.
Trends in atmospheric carbon dioxide, 2026.
Additional data accessible via Scripps Institution of Oceanography: https://scrippsco2.ucsd.edu/.

Martin Leduc, Alain Mailhot, Anne Frigon, Jean-Luc Martel, Ralf Ludwig, Gilbert B Brietzke, Michel Giguère, François Brissette, Richard Turcotte, Marco Braun, et al.
The climex project: A 50-member ensemble of climate change projections at 12-km resolution over europe and northeastern north america with the canadian regional climate model (crcm5).
Journal of Applied Meteorology and Climatology, 58(4):663–693, 2019.

References II

Felix Schur and Jonas Peters.
Decor: Deconfounding time series with robust regression, 2024.

Sebastian Sippel, Nicolai Meinshausen, Anna Merrifield, Flavio Lehner, Angeline G. Pendergrass, Erich Fischer, and Reto Knutti.
Uncovering the forced climate response from a single ensemble member using statistical learning.
Journal of Climate, 32(17):5677 – 5699, 2019.

Patrick Zippenfenig.
Open-Meteo.com Weather API.

Backup

Proofs - Torrent (probabilistic)

Define the square root of minimum eigenvalue of all possible clean subsets as

$$\lambda_{a_n}(X) := \min_{S \subseteq \{1,\ldots,n\} \text{ s.t. } |S|=a_n} \sqrt{\lambda_{\min}(X_S^\top X_S)}$$

Guarantee of Torrent with sub-Gaussian noise (Corollary 4)

Let ϵ be i.i.d. zero-mean sub-Gaussian noise with variance proxy σ^2 and assume $c_n < n/2$. Then, there exists a constant $K > 0$ s.t. $\forall\ \delta > 0$ with probability $\geq 1 - \delta$:

$$\begin{aligned}\|\hat{\beta}_{\text{Tor}}^{n,a_n} - \beta\|_2 \leq \frac{\sigma}{\lambda_{a_n}(X)} \left(1 + \frac{1}{1 - \sqrt{2}\eta}\right) \left(\sqrt{d} + \sqrt{2c_n} L_{n,\delta}\right) \\ + \frac{2\sigma\sqrt{c_n}\,(1 + L_{n,\delta})}{\lambda_{a_n}(X)(1 - \sqrt{2}\eta)}.\end{aligned}$$

with $L_{n,\delta} := \sqrt{K \log\left(2en/c_n\delta\right)}$ representing the logarithmic search penalty.

Insights:

- What happens if $c_n = 0$? Compare it to standard OLS.
- If the rows of X are i.i.d. standard Gaussian random vectors (i.e. $\lambda_{a_n}(X) \in \Omega_{\mathbb{P}}(\sqrt{n})$ [1]) and $c_n \in o(n/\log(n))$, then $\hat{\beta}_{\text{Tor}}^{n,a_n}$ is consistent.

Convergence properties of DecoR-Tor (Theorem 6) - shortened

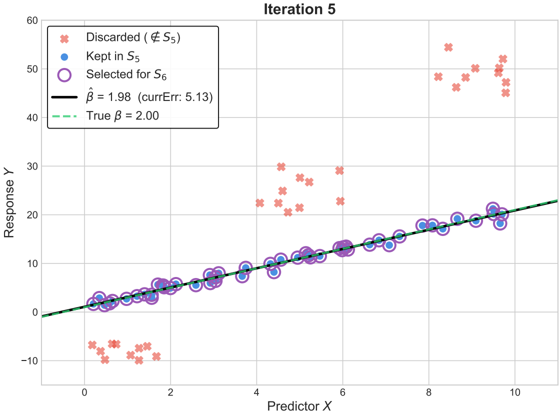
$$\|\hat{\beta}^{\phi,n}_{\mathrm{DecoR}} - \beta\|_2 \in \mathcal{O}_{\mathbb{P}}\left(\sigma\sqrt{\frac{c_n \log(n/c_n)}{n}}\right).$$

Proof ideas:

- By some assumptions, note that $\sqrt{\lambda_{\min}\left((X^n_\phi)^\top_S (X^n_\phi)_S\right)} \geq c'$ with high probability.
- each component of $\eta^n_\phi = \{T^{\phi,n}_k(\eta)\}_{k\leq n} = \frac{1}{n}\sum_{l=1}^n (\eta_{Tl/n})_k \cdot \phi_k(Tl/n)$ has (by factor n) reduced variance $\bar{\sigma}^2 = \sigma^2/n$
- Use **Corollary 4** and derive Big-$\mathcal{O}$ result:

$$\|\hat{\beta}^{n,a_n}_{\mathrm{Tor}} - \beta\|_2 \leq \frac{\sigma}{\lambda_{a_n}(X)}\left(1 + \frac{1}{1-\sqrt{2}\eta}\right)\left(\sqrt{d} + \sqrt{2c_n K \log\left(2en/c_n\delta\right)}\right) + \frac{2\sigma\sqrt{c_n}\left(1+\sqrt{K\log\left(2en/c_n\delta\right)}\right)}{\lambda_{a_n}(X)(1-\sqrt{2}\eta)}.$$

Torrent Algorithm [1] - Iteration-wise Visualization

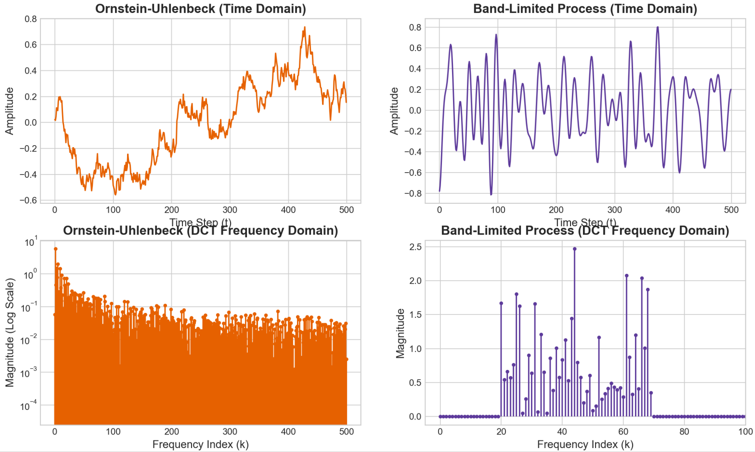


Using $n = 80, a = 50, c = 30$

Start with:

$t = 5, \text{currErr} = 5.13$

Visualize OU and BLP



Simulation Parameters: Number of data points (n) = 500. Ornstein-Uhlenbeck AR coefficient (φ) = 0.9984. Band-Limited Process active frequency band = [20, 70).